

Modeling the Hydropower System of the River Ångermanälven in Spine

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Abstract—As the demand for electricity continues to increase, power generation must scale accordingly while minimizing additional environmental impact. One approach to achieving this is through the expanded implementation of solar and wind power generation. However, due to the intermittent nature of these energy sources, controllable backup systems are required during periods of low production. This creates a need for a balance between power grid expansion and hydropower simulation to ensure stable and reliable grid operation. This study therefore aims to provide a comprehensive guide for modeling a complex hydropower system. The open-source program Spine Toolbox together with SpineOpt were used to model the hydropower system of the Ångermanälven, as they provide the foundation of an already capable energy system simulation framework. In addition, a structured Excel based setup was developed to simplify the adjustment and import of model parameters. The resulting model successfully interconnects and simulates all power plants with realistic water flows, including effects such as spillage, environmental inflow, and real world electricity prices. Although the model does not yet achieve full real-world accuracy, it provides a solid foundation for further development and represents a useful step toward supporting future environmental and energy goals.

Sammanfattning—I takt med att efterfrågan på elektricitet fortsätter att öka måste elproduktionen skalas upp samtidigt som ytterligare miljöpåverkan minimeras. Ett sätt att uppnå detta är genom en utökad implementering av sol- och vindkraft. På grund av dessa energikällors intermittenta natur krävs dock reglerbara reservsystem under perioder med låg produktion. Detta skapar ett behov av balans mellan utbyggnad av elnätet och simulering av vattenkraft för att säkerställa en stabil och tillförlitlig drift av kraftsystemet. Denna studie syftar därför till att tillhandahålla en omfattande guide för modellering av ett komplext vattenkraftssystem. Det öppna källkods verktyget Spine Toolbox tillsammans med SpineOpt användes för att modellera vattenkraftssystemet i Ångermanälven, då de utgör grunden för ett redan kapabelt verktyg för energisystemsmodellering. Därutöver utvecklades en strukturerad Excel baserad lösning för att förenkla justering och import av modellparametrar. Den resulterande modellen sammankopplar och simulerar samtliga kraftverk med realistiska vattenflöden, inklusive effekter såsom spillvatten, miljötillrinning och verkliga elpriser. Även om modellen ännu inte uppnår fullständig verklighetstrogn noggrannhet utgör den en stabil grund för vidare utveckling och representerar ett användbart steg mot att stödja framtida miljö- och energimål.

Index Terms—Hydropower, Simulation, Modeling, Sweden, SpineToolbox, SpineOpt, Ångermanälven, Balance

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I. INTRODUCTION

Electricity is fundamental to modern society, and global demand continues to increase. Meeting this demand sustainably

remains a central challenge in the energy transition. One solution is hydropower which currently accounts for approximately 20% of global electricity production, owing to its operational flexibility, low operating costs, and independence from fuel market price fluctuations [1].

Within the European Union, climate policy has progressively strengthened renewable energy targets. Since the first renewable energy directive in 1997, targets have been expanded, and in 2021 the renewable energy goal was increased to 42.5% by 2030 [2]. Furthermore, the European Union has committed to achieving climate neutrality by 2050 [3]. In parallel, environmental regulations have tightened, including strengthened requirements for ecological water flow and reduced downstream environmental impact [4].

Sweden represents a particularly relevant case study. The country has already achieved a renewable energy share of 62.8% and has set a national climate neutrality target for 2045, five years ahead of the European Union's objective [2]. Hydropower constitutes approximately 54% of Sweden's renewable electricity generation [5], making it central to both national and European climate strategies.

Since hydropower in Sweden is already largely developed, future improvements are more likely to come from modernization and increased operational flexibility rather than the construction of new power plants. As the share of intermittent renewable energy sources such as solar and wind power continues to increase, the demand for balancing and energy storage within the power grid also grows. In this context, hydropower plays an important role due to its fast response times, controllability and inherent storage capacity. Recent research highlights that refurbishing and upgrading existing hydropower facilities can significantly improve generation flexibility and regulation capability while minimizing additional environmental impact compared to new large scale developments. Therefore, improving the efficiency and operational adaptability of existing hydropower systems may become an essential step toward supporting a stable and reliable renewable based energy system [6].

Calculating these balancing requirements prior to development will require realistic and dependable simulations of different hydropower systems. In particular, the Swedish Ångermanälven river system represents a relevant case study, as it constitutes the most complex hydropower system in the country. Although previous studies have examined the river system, their focus has differed, primarily addressing the environmental impact of the system and the development of a partially functional simulation model [7], [8]. In contrast, this study aims to provide a comprehensive guide for modeling a complete hydropower simulation. The analysis focuses on

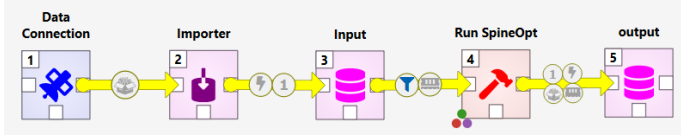


Fig. 1. The five main blocks that make up a complete hydropower simulation



Fig. 2. The four main classes in Spine

37 hydropower plants within the Ångermanälven river system, including facilities located along the main river itself and its tributaries, Faxälven, Fjällsjöälven, and Ledingsån. A complete list of the studied plants is provided in the Appendix. All simulation files used in this study are available at: <https://github.com/AralTurkmen/Bachelor-Thesis>

II. SPINE FRAMEWORK

In order to simulate a large hydroelectric power system, the open-source project Spine is used. Spine is specifically developed for energy system modeling and has been supported through EU funded initiatives, aligning with the objectives of the new directive [9]. The Spine ecosystem consists of several components that address different aspects of energy system simulations. The primary component used in this study is Spine Toolbox, a graphical data source and workflow management tool that facilitates model configuration, execution, and data handling [10]. In addition, SpineOpt is employed in combination with Spine Toolbox to provide advanced optimization capabilities and enable detailed modeling of complex energy systems, including hydroelectric power systems [11].

To build a functional hydropower model in Spine, the simulation relies on five main blocks, as illustrated in Fig. 1. The first block, “Data Connection,” links to external data sources, which in this case is an .xlsx file. The “Importer” block defines how this Excel file is read, typically sheet by sheet. This setup makes it easier to handle and import large sets of parameters, where each sheet contains a predefined group of sub parameters. The “Input” block serves as the core environment where all necessary classes, objects, and relationships required for the “SpineOpt” block are created. Within this block, data can either be entered manually or imported directly from the “Importer” block. It is important to note that everything in the “Input” block is case sensitive, meaning even small errors, such as incorrect capitalization, can cause the simulation to fail. The “SpineOpt” block is responsible for running the simulation using the defined classes and parameters and typically does not require any modification from the user. Finally, the “Output” block presents the results by plotting the selected output parameters defined in the “Input” block, within the specified simulation boundaries.

The core simulation model is built upon four main classes, as illustrated in Fig. 2. The “node” class acts as a reservoir for the commodity and includes parameters that define how much additional water enters from tributaries, which will be described further later in the report. The “unit” class functions as the generator in the model, where water flow is converted into energy. The “connection” class is responsible for directing how and where the commodity flows, in this case both water and electricity. Finally, the “commodity” class is used to define the different types of commodities [9].

III. CASE STUDY OF ÅNGERMANÄLVEN

This section presents a detailed description of how a hydropower system can be modeled in Spine Toolbox. The work is based on readily available, although dated, online tutorials, with the objective of fully modeling the Ångermanälven river system while discussing the challenges and errors encountered during the process. Finally, the automation of model building will be explained, and the results produced by the model will be examined.

A. Small-scale modeling

To start building the river system model, a small-scale version of three power plants is preferred in order to more easily identify issues and add new capabilities to the model. To achieve this, the SpineOpt tutorial “How to model hydropower coupling” [9] is followed. This tutorial supports the development of an initial model with two power plants. The model is capable of handling water flow, reservoirs, and electricity generation through discharge, as well as spillage when reservoirs exceed their capacity.

It is useful to be aware of the tutorial’s flaws. To simplify the process of setting up a first model, a template can be downloaded from the same GitHub page under “templates” [12]. As of the time of writing, the template is missing a required parameter, which causes a load error. In this case, the missing parameter was “use_economic_representation” with the error displayed in the Julia console. To resolve this issue, the user must manually add the parameter under the “Parameter definition” tab in the Spine DB Editor, which is the editor used to define and build the model. This process may need to be repeated for several parameters. Once no further load errors are displayed, the issue can be considered resolved.

Another flaw is that the tutorial does not mention one relationship class, “connection__to_node”, which ensures that water flows from the lower reservoir of one power plant to the upper reservoir of the next, as shown in Fig. 3. Simply adding the mentioned relationship class together with the parameter “fix_ratio_out_in_connection_flow” set to a value of “1” will fix this issue.

All the non-constant parameters can be seen in Table I. These parameters are valued after each specific river, meaning that other river systems will require a new set of parameter values, which may differ significantly from those used in this case study.

SpineOpt uses these parameters to determine the most efficient way to operate the hydropower plants. A negative

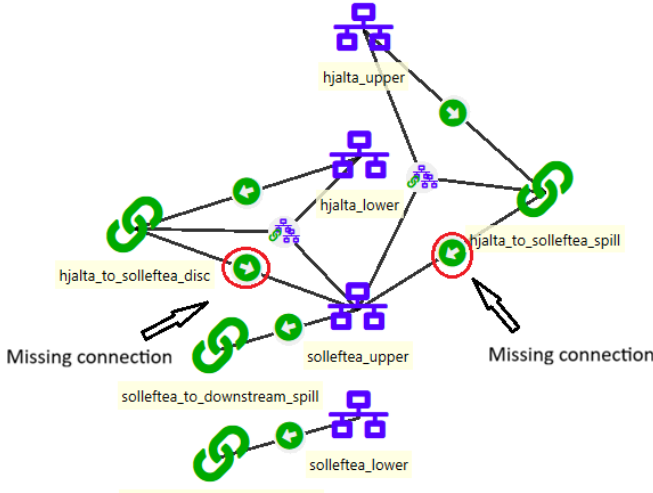


Fig. 3. Connections between nodes depicting water flow

TABLE I
NON-CONSTANT OPERATING PARAMETERS IN THE MODEL

Class in spine	Parameter in spine	Definition
connection__node__node	connection_flow_delay	Travel time for water between plants
node	demand	Natural inflow of water
node	fix_node_state	Initial and final reservoir levels
node	node_state_cap	Maximum capacity of the reservoir
unit__from__node	unit_capacity	Maximum water flow through turbine
unit__from__node	vom_cost	Electricity prices
unit__node__node	fix_ratio_out_in_unit_flow	Conversion rate from water to electricity
unit__to__node	unit_capacity	Maximum installed power in the plant

demand value is assigned to the upper reservoir to represent water flowing in from upstream. This method can be applied to all nodes and allows real-world data, such as rainfall and tributary inflows, to be imported and used to simulate how water flow changes between plants.

The "fix_node_state" uses a time series to set the reservoir level at the first and second to last time steps to a chosen value. In later, more realistic models, this can be replaced by importing actual reservoir level data for the period being simulated. Time series are also used when importing electricity prices.

B. Full-scale modeling

Using the experience and understanding of the program limitations from the small scale model, a full scale model can now be created. However, at this point it is both risky and time

consuming to continue using the Spine DB Editor in order to define new parameters and class objects. Instead Excel can be used to import all parameters and classes by duplicating names to minimize the risk for spelling mistakes. The use of Excel and the associated import tool will be covered in the following subsections, as the procedure can be somewhat finicky and requires careful handling. Note that just as the small-scale model, this large-scale model is also limited to electricity prices, water flow represented as negative demand, the efficiency and capacity of the plants.

In order to use Excel and create a realistic model, representative data must be used. A large amount of relevant data is publicly available through previous studies and various online sources. One particularly useful source was vattenkraft.info, which was crucial in providing a clearer understanding of the river system's geographical scale and overall dynamics [13]. The main operating parameters for all hydropower plants in Ångermanälven were obtained from a previous master's thesis [14]. Finally, the electricity price data was obtained from Ember Energy [15]. The dataset included electricity prices dating back to 2015 in three different temporal resolutions: hourly, daily, and monthly electricity prices in euros per megawatt-hour (EUR/MWh). To reduce both simulation and import times, the model used an hourly resolution for the first week of 2015.

C. Excel configuration

Before configuring the import module in Spine, the Excel document has to be created. A simple rule to keep in mind is that only objects and relations that share the same number of dimensions can be placed in the same sheet. This follows from the relatively simple reading rules in Spine, where each row or column is assigned to a specific parameter.

For good practice and ease of use, every Excel sheet should start with a header similar to Fig. 4, as this makes the importer easier to configure later on. One of the main challenges is the case sensitivity in Spine. This can result in Excel blocks unintentionally sharing common sources, since they reference parameters defined earlier in other sheets.

Another major challenge during the data import was handling time series. When extracting a time series as a string from Spine, it is for example given in the following .JSON format:

```
{
  "data": {
    "2015-01-01T00:00:00": -23.37,
    "2015-01-01T01:00:00": -19.33,
    "2015-01-01T02:00:00": -17.66,
    "2015-01-01T03:00:00": -17.53,
    "type": "time_series"
  }
}
```

This format can be generated in Excel using the following expression:

```
={"data": {" & TEXTJOIN(", ";
SANT;E2:E1000) & "}, "type":
"time_series"}
```

Here, "E2" to "E1000" refer to cells in Excel containing data in the ISO 8601 format with Swedish abbreviations:

class name	unit	node	node	parameter name	alternative	parameter value
unit_node_node	stalón_plant	stalón_lower	stalón_upper	fix_ratio_out_in_unit_flow	Base	1
unit_node_node	malgomaj_plant	malgomaj_lower	malgomaj_upper	fix_ratio_out_in_unit_flow	Base	1
unit_node_node	volgsjöfors_plant	volgsjöfors_lower	volgsjöfors_upper	fix_ratio_out_in_unit_flow	Base	1
unit_node_node	stenkullafors_plant	stenkullafors_lower	stenkullafors_upper	fix_ratio_out_in_unit_flow	Base	1
unit_node_node	asele_plant	asele_lower	asele_upper	fix_ratio_out_in_unit_flow	Base	1
unit_node_node	hallby_plant	hallby_lower	hallby_upper	fix_ratio_out_in_unit_flow	Base	1
unit_node_node	gulsele_plant	gulsele_lower	gulsele_upper	fix_ratio_out_in_unit_flow	Base	1
unit_node_node	degerforsen_plant	degerforsen_lower	degerforsen_upper	fix_ratio_out_in_unit_flow	Base	1
unit_node_node	edenforsen_plant	edenforsen_lower	edenforsen_upper	fix_ratio_out_in_unit_flow	Base	1
unit_node_node	langbjörn_plant	langbjörn_lower	langbjörn_upper	fix_ratio_out_in_unit_flow	Base	1

Fig. 4. Shows a small part of the excel sheet containing a handfull of the unit_node_node relationship class objects, with their parameters defined in the header

"åååå-MM-ddTtt:mm:ss"

One limitation of using the expression is that it is stored within a single Excel cell. This quickly leads to the cell character limit being exceeded, resulting in a calculation overflow error. Through testing, the maximum number of allowed entries was found to be approximately 1095, corresponding to roughly 45 days of hourly resolution data. To overcome this limitation, the data can instead be aggregated into daily or monthly mean values, thereby enabling simulations over longer time periods. However, the use of mean values only provides a rough approximation of the actual system behavior. Additionally, if a time series contains a data point with "NaN", all other values must be of integer type, otherwise the simulation fails. Electricity price data was initially imported from a CSV file into Excel and subsequently reduced into usable subsets for different temporal resolutions, including hourly, monthly, and annual data. These subsets were then combined into the previously described expression.

D. Data import

Configuring the importer tool is quite intuitive once the user becomes familiar with the interface, shown in Fig. 5. On the left-hand side of the window, the user can choose which Excel sheets to import by checking their respective boxes. Note that the Excel file cannot be open at the same time, as this blocks Spines read access to the document.

Moving to the middle section, there are two noteworthy parts. The top part contains the sheet rules, where one important option must be enabled in this case: the "Has header:" option. This ensures that Spine does not interpret the first header row in the Excel sheet as data and makes it easier to configure the reading rules. Below these settings, there is a sheet preview that shows how Spine reads the data. Initially, the entire sheet will appear white, as no reading rules have been defined yet. The header row in the preview should contain the "A" character in every column, this ensures that all data is read as strings, making it compatible with the data in the Excel document.

On the right-hand side of the window, the reading rules are configured. At the top, there is a mappings section, which acts as presets for reading rules if the user wants to simulate different scenarios. The section below is used to configure

the reading rules themselves. All imported sheets are treated as "Entity class" types, since both relations and objects are based on classes.

Regarding the parameter type, if the sheet is only used to initialize classes such as node, unit, connections and commodities with their respective names, this can be set to "None." If the sheet contains relation parameters, the parameter type should instead be set to "Value," followed by selecting "Single value." The number of dimensions depends on how many objects are involved: during class initialization, only one is needed, whereas relations typically require two or three. Since the Excel sheets include headers, the dimension name categories can easily be imported by linking them to their respective column headers. Finally, the remaining categories are assigned to their corresponding columns.

E. Results

The large-scale model is able to handle water flow between power plants and nodes with a time delay, simulating the time it takes for water to move downstream. It can model power generation by converting water into electricity and "selling" it on a fixed energy market based on 2015 prices. It also accounts for spill when incoming water exceeds turbine capacity, allows the user to specify the starting and ending reservoir water levels, and simulates the difference in incoming water flow to the plants due to tributary inflow.

Looking at Fig. 6 - 9 a small part of the complete system can be seen working. The Bågede hydropower plant is ten hours up streams from Lövön and both of them have a constant discharge of water through them. Note that the reservoir levels are in hour equivalents, which correspond to the volume represented by a flow of $1 \text{ m}^3/\text{s}$ during one hour, equivalent to $3,600 \text{ m}^3$ or 3.6 million liters per hour. This is done to facilitate the conversion between water flow and volume, and then back to flow.

From Fig. 10, which shows the total power generation of all power plants, the system mostly operates at maximum capacity. However, during certain hours when the electricity price is at its lowest, as seen in Fig. 11, production decreases.

IV. DISCUSSION

Looking at the results in Fig. 10 and Fig. 11, it becomes clear that the model is able to respond to changes in electricity

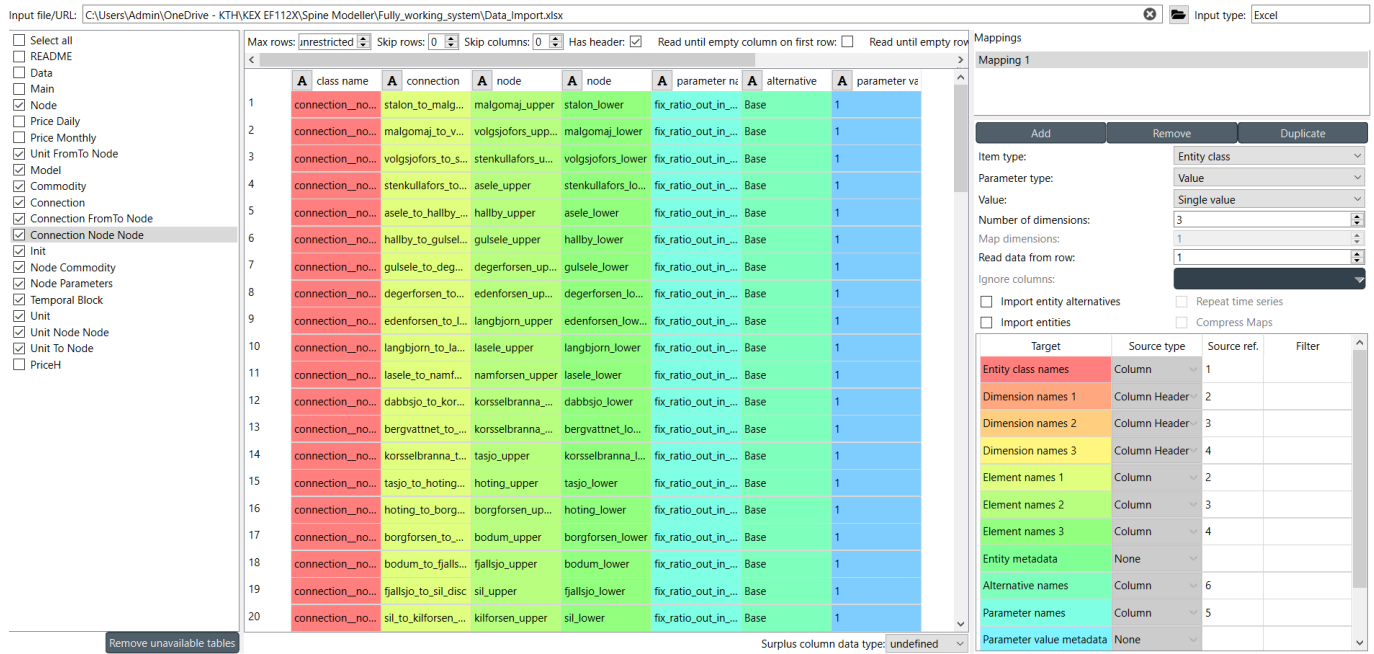


Fig. 5. Overview of the importer tool in Spine

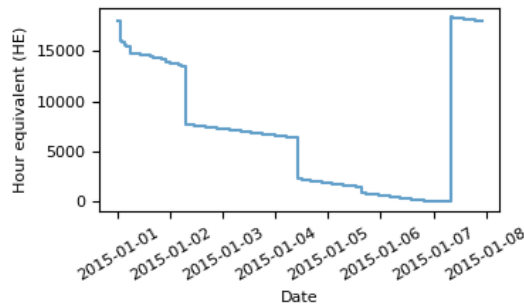


Fig. 6. Reservoir level for Bågede power plant

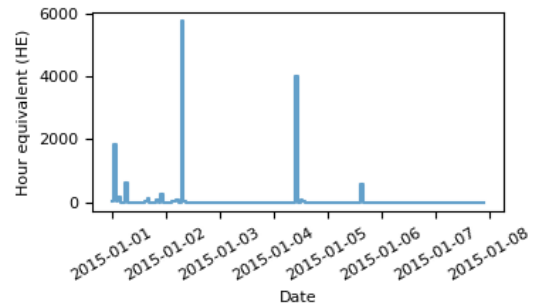


Fig. 8. Water spilled from the Bågede reservoir

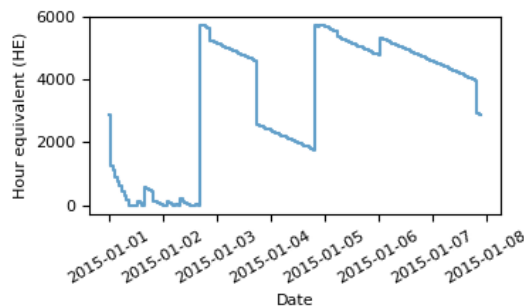


Fig. 7. Reservoir level for Lövön power plant

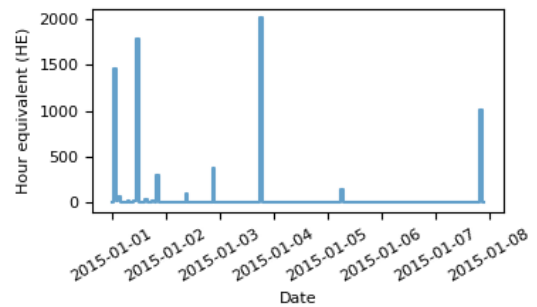


Fig. 9. Water spilled from the Lövön reservoir

prices by adjusting the total energy production of the hydropower plants. This can also be interpreted as the production balancing peak demand in the system, which matches the goal of this study: to simulate the balancing role of hydropower in an electricity grid with mainly weather dependent power

production, supporting a greener energy transition.

From Fig. 11, it can also be observed that the hydropower plants often operate at maximum capacity. Earlier studies have shown that in a more accurate simulation, the production typically follows the electricity price more closely, with maximum

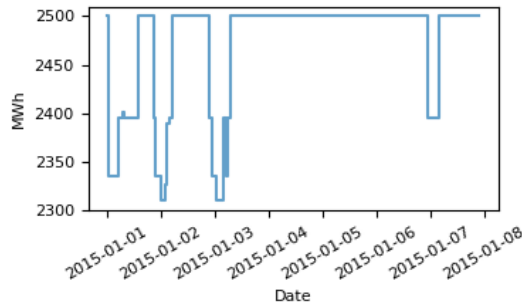


Fig. 10. The total energy production of all power plants

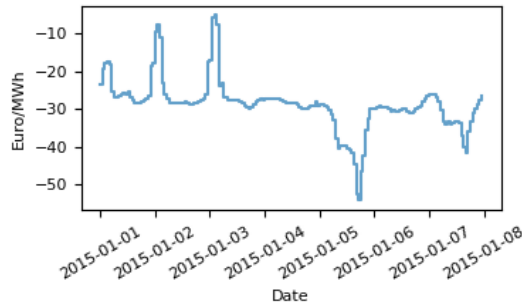


Fig. 11. The electricity price, sign inverted

production occurring only during short periods of very high prices [7]. During periods of low electricity prices, most plants usually stop producing entirely, resulting in a significantly lower total production than what is shown in Fig. 11.

It is difficult to obtain definitive answers about how the river system actually operates in reality, since the output depends on more factors than our current large-scale model includes. The model needs to incorporate more varied tributary inflows and total water flow between the plants depending on the season, simulate power plant efficiency with greater accuracy, and include a minimum spillage requirement for environmental considerations.

From the simulation results, by examining two power plants, Bågede and Lövön, we can see how the model operates. In Fig. 8, two spill events appear a few hours after January 2nd and around 12 PM on January 4th. According to the provided Excel sheet, it takes ten hours for the spill to reach the Lövön reservoir. Ten hours later, in Fig. 7, two corresponding spikes can be observed. This shows that the system runs with a realistic water flow between the power plants and reservoirs.

The same graphs also highlight an improvement that could be made in a future project. Both Bågede and Lövön respectively start and end with the same amount of stored water. This is because the current model assigns no value for storing water for future use. Therefore, the model forces the water level to return to a fixed volume at the end of the simulation, otherwise it could result in the reservoirs being completely emptied. This is important to ensure that no reservoir becomes prematurely depleted over the course of the year. However, this behavior could be improved by incorporating a predicted

future electricity price beyond the span of the simulation, making the model prefer storing water over selling during periods with a low electricity price.

Finally, the import automation blocks were shown to be necessary for accurately constructing a complex model such as the selected case study. However, it also highlights the potential of this approach, where the customizability of Excel facilitates adaptation to other, and potentially even more complex, hydropower systems. Without requiring extensive knowledge of the underlying program structure, such as the class relationships.

V. CONCLUSION

A. Conclusion of the Study

The study set out to better understand how an optimal hydropower system can be modeled to balance a weather-dependent electricity grid. The Spine program was found to effectively simulate water and energy markets, optimizing for efficiency and profit. It follows electricity prices, demonstrating that hydropower is effective in regulating large variations in energy demand. This makes it a promising tool for grid balancing in the expansion of solar and wind power generation.

However, its regulating ability during periods of maximum production becomes limited, since the plants cannot further increase production during even higher electricity prices. This limitation is not caused by insufficient turbine capacity, but rather by a lack of realistic tributary and seasonal data, where significantly more water enters the system than would occur under real-world conditions.

In order to achieve good results the model requires a wide range of real-world data, which can limit projects where such data is unavailable or difficult to obtain. Even with appropriate data, the current model still lacks some realism and has several potential areas for improvement.

With enough data, however, despite its flaws and the assumed ideal scenario mentioned above, it remains a strong tool for simulating reality to a reasonable extent without requiring excessive details. By finding an optimal solution, it can be used to identify the most effective set of adjustable parameters to improve efficiency for the simulated system.

B. Future Work

First, future work should build upon this study to gain traction quickly, as it currently serves as the most up to date reference available. Moving forward, the main focus should be on expanding the simulation parameters to better reflect real-world conditions. This includes the use of realistic turbine operating curves, introducing a minimum spillage requirement to account for environmental considerations, allowing water flow to vary with the seasons, and incorporating predicted future electricity prices so that the model values storing water for later use. Another important step would be to increase the capacity of the current model, for example, by using alternative file formats to avoid Excel's calculation overflows when importing large time series datasets.

APPENDIX

TABLE II
HYDROPOWER PLANTS AND RESPECTIVE WATERCOURSES

Power-plant	Watercourse
Bergvattnet	Fjällsjöälven
Blåsjön	Faxälven
Bodum	Fjällsjöälven
Borgforsen	Fjällsjöälven
Bågede	Faxälven
Dabbsjö	Fjällsjöälven
Degerforsen	Ångermanälven
Edensforsen	Ångermanälven
Edsele	Faxälven
Fjällsjö	Fjällsjöälven
Forsmo	Ångermanälven
Forsse	Faxälven
Gulsele	Ångermanälven
Gäddede	Faxälven
Hällby	Ångermanälven
Hjälta	Faxälven
Hoting	Fjällsjöälven
Junsterforsen	Faxälven
Kilforsen	Fjällsjöälven
Korsselebränna	Fjällsjöälven
Långbjörn	Ångermanälven
Lasele	Ångermanälven
Ledinge	Ledingsån
Linnvasselv	Faxälven
Lövön	Faxälven
Malgomaj	Ångermanälven
Moforsen	Ångermanälven
Nämforsen	Ångermanälven
Ramsele	Faxälven
Sil	Fjällsjöälven
Sollefteå	Ångermanälven
Stalon	Ångermanälven
Stenkullafors	Ångermanälven
Storfinnforsen	Faxälven
Tåsjö	Fjällsjöälven
Volgsjöfors	Ångermanälven
Åsele	Ångermanälven

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